

CONSTRUCTEDGE PROCUREMENT MIDTERM

Data Overview & Quality Report

Final instructor-approved analysis dataset

— VALIDATED DECISION BRIEF —

23 July 2026

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1. Dataset Overview

The official analysis dataset contains 400 distinct transactions from 2025 covering 10 projects, 24 suppliers, six material categories, and five procurement methods. After the instructor-approved identifier corrections, the dataset contains 400 unique PO_Number values. It tracks budget, purchase-order value, cost variance, supplier, material category, procurement method, lead time, and delivery outcome.

Measure	Validated Result	Comment
Rows / transactions	400	All retained
Unique PO numbers	400	Two approved identifier corrections applied
Projects	10	Unchanged
Suppliers	24	Unchanged
Material categories	6	Unchanged
Procurement methods	5	Invitational is not observed
Total budget	\$431,518,839.83	Authoritative total
Total PO value	\$418,104,662.98	Reconciled total
Total cost variance	-\$13,414,176.85	PO value – budget

2. Instructor-Approved Cleaning Actions

Field	Original Issue	Final Action	Validation
PO_Number	Two duplicate identifiers	PO-2025-0005 → PO-2025-0030 (row 31); PO-2025-0060 → PO-2025-0120 (row 121)	PASS — 400 unique PO values
PO_Date	Four date formats	Converted all values to true dates displayed as yyyy-mm-dd	PASS — 400/400 valid dates
Month	Two date/month disagreements	Regenerated Month from PO_Date at rows 91 and 201	PASS — 400/400 matches
Unit_Rate_CAD	Two missing rates	Derived PO_Value_CAD ÷ Quantity at rows 17 and 202	PASS — both reproduce PO value
PO_Value_CAD / Unit_Rate_CAD	Two financial identity failures	At PO-2025-0045 and PO-2025-0223, Budget Value and Cost Variance are authoritative; PO Value = Budget + Variance, then Unit Rate = PO Value ÷ Quantity	PASS — 400/400 identities
Lead_Time_Days	Two missing Civil Works values	Imputed instructor-approved category median of 45 days	PASS — no ranking or recommendation change

The complete cell-level Audit_Log contains 412 records: 400 date standardizations, four deterministic corrections, six derived values, and two imputations. Every record includes the original and cleaned value, action type, justification, validation result, confidence level, and Instructor Approved status.

3. Validation Outcome

Acceptance Area	Result
Row, project, supplier, category, and method counts	PASS
Budget, PO value, and cost variance totals	PASS
Quantity × Unit Rate = PO Value	PASS — 400/400
Quantity × Budgeted Rate = Budget Value	PASS — 400/400
PO Value – Budget Value = Cost Variance	PASS — 400/400
Variance_% recalculation	PASS — 400/400
Date and Month consistency	PASS — 400/400
Missing-value counts	PASS — 0 remaining in analysis fields
Q1-Q4 two-method verification	PASS — 83/83 MATCH
Final workbook QA	PASS — 37/37

Analysis readiness: FULLY ANALYSIS-READY. All instructor-approved rules are applied, all 31/31 acceptance tests pass, and no instructor decisions remain unresolved.

4. Analysis Rules Used in Q1-Q5

- Q1 includes all 400 records. Positive variance means over budget; negative means under budget. Net percentage uses aggregate variance divided by aggregate budget.
- Q2 defines “consistently above budget” as more than 50% of a supplier’s orders having positive variance and the supplier’s total variance being positive.
- Q3 compares the five observed procurement methods with descriptive statistics and row-normalized tests. Invitational has no observed records.
- Q4 includes Delivered orders only. Pending, Cancelled, and Partially Delivered orders are excluded.
- Q5 recommendations are evidence-linked scenarios; financial impact estimates overlap and are not additive.

5. Final Tool Record

Deliverable Stage	Tool / Output
Data cleaning and validation	Excel-compatible audited workbook, Python, independent pivot/groupby logic
Statistical analysis	Python/SciPy and independent aggregation
Executive report	Microsoft Word-compatible DOCX and verified PDF
Interactive dashboard	Self-contained HTML and hosted web application
Presentation	PowerPoint-compatible PPTX
Website content	Four-page Wix-ready copy: Home, Data Analysis, Materials, Contact

Release status: Fully analysis-ready. All listed outputs use the same validated release dated 23 July 2026.